

ROSETOWN EAST, SK, Canada

WMO: 715100

Lat: 51.570N

Lon: 107.920W

Elev: 1923

StdP: 13.70

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.3	-22.4	-35.2	0.8	-28.1	-28.8	1.2	-22.4	28.9	26.6	25.7	20.1	6.8	260	0.717

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.1	87.6	63.7	83.7	62.6	80.3	61.8	68.6	79.5	66.3	77.4	64.3	75.6	11.2	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.0	99.3	73.9	62.5	90.8	71.1	60.2	83.6	68.7	34.0	79.5	32.1	77.3	30.5	75.7	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	22.8	20.2	-34.6	94.4	5.7	3.8	-38.7	97.1	-42.1	99.4	-45.3	101.5	-49.5	104.3
			-34.7	73.9	5.6	3.3	-38.7	76.3	-42.0	78.2	-45.1	80.1	-49.1	82.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.9	6.1	9.0	21.8	38.9	51.0	59.7	64.3	62.8	53.7	39.4	23.2	10.7
	DBStd	23.64	15.74	14.36	14.58	10.20	8.29	6.36	5.49	6.30	8.43	9.13	12.37	14.31
	HDD50	6246	1360	1147	873	348	89	5	0	1	54	345	805	1219
	HDD65	10411	1825	1567	1338	782	437	182	79	119	349	794	1255	1684
	CDD50	1456	0	0	0	16	121	296	445	396	165	17	0	0
	CDD65	144	0	0	0	0	4	22	59	49	10	0	0	0
	CDH74	2658	0	0	0	15	180	386	848	895	324	10	0	0
	CDH80	813	0	0	0	2	38	99	253	312	108	1	0	0
Wind	WSAvg	9.4	9.2	8.9	9.9	10.8	11.0	10.0	8.1	8.1	9.1	9.6	9.2	8.7
Precipitation	PrecAvg	14.8	0.7	0.4	0.6	1.0	1.7	3.1	2.5	1.6	1.1	0.7	0.6	0.5
	PrecMax	29.8	2.0	0.9	1.5	5.2	4.7	5.9	6.7	3.9	3.1	2.1	1.6	1.0
	PrecMin	6.5	0.1	0.1	0.1	0.0	0.2	0.7	0.6	0.0	0.0	0.2	0.0	0.1
	PrecStd	4.7	0.4	0.2	0.4	1.0	1.2	1.4	1.6	0.9	0.9	0.5	0.4	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.3	39.9	57.4	75.2	84.9	88.4	91.6	92.8	89.1	74.3	57.4	42.8
		MCWB	36.4	36.1	45.6	52.5	58.2	63.4	66.5	64.5	61.6	54.9	46.6	37.4
	2%	DB	37.0	36.2	49.9	68.4	79.2	82.4	86.2	87.7	82.9	66.1	48.6	38.5
		MCWB	34.2	33.3	42.5	49.6	55.7	61.1	66.3	62.8	58.8	50.6	41.5	34.7
	5%	DB	33.3	33.1	43.8	62.7	74.2	78.4	82.1	83.2	77.0	59.9	43.2	34.8
		MCWB	31.2	31.3	38.6	47.1	54.1	59.7	65.5	62.8	56.7	47.7	37.8	32.0
	10%	DB	28.5	29.7	38.7	57.0	69.5	74.5	78.6	79.0	71.3	54.7	38.9	30.3
		MCWB	27.1	28.2	35.6	44.2	51.8	58.1	64.4	61.8	54.4	45.2	35.4	28.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.8	36.2	47.2	54.1	60.8	68.3	72.8	69.9	64.5	56.8	47.4	37.9
		MCDB	39.5	39.2	56.2	71.5	76.8	79.1	83.2	81.6	84.0	68.6	56.4	41.8
	2%	WB	34.4	33.6	42.4	50.7	57.9	64.7	69.8	67.1	60.6	51.8	41.9	34.9
		MCDB	36.9	35.8	49.0	65.2	74.5	76.3	80.5	79.1	77.8	62.4	48.1	38.3
	5%	WB	31.3	31.2	38.8	48.0	55.5	62.2	67.6	65.0	57.8	48.7	38.4	32.0
		MCDB	33.3	33.1	43.8	59.8	70.9	72.9	77.8	77.6	73.3	58.3	42.3	34.7
	10%	WB	27.1	28.2	35.7	45.3	53.1	60.2	65.5	63.0	55.5	46.0	35.5	28.4
		MCDB	28.5	29.6	38.5	55.7	67.2	70.6	75.7	75.2	69.5	54.2	38.7	30.2
Mean Daily Temperature Range	5% DB	MDBR	18.1	19.2	18.3	23.2	26.0	22.2	24.1	26.2	26.2	22.8	17.8	17.3
		MCDBR	20.2	21.0	23.0	33.9	34.7	29.9	30.7	34.9	37.6	32.1	24.0	20.9
		MCWBR	18.3	18.9	16.0	18.1	16.6	13.5	14.9	15.0	16.8	18.4	17.2	17.5
	5% WB	MCDBR	20.1	20.7	22.0	30.5	29.9	24.0	26.6	28.3	32.9	29.7	22.3	20.9
		MCWBR	18.4	18.8	15.6	17.2	15.1	12.3	14.8	14.3	15.6	17.8	16.4	17.9
Clear-Sky Solar Irradiance	taub	0.264	0.266	0.285	0.349	0.370	0.373	0.377	0.386	0.333	0.289	0.260	0.240	
	taud	2.281	2.344	2.448	2.310	2.299	2.346	2.356	2.297	2.425	2.529	2.474	2.380	
	Ebn at Noon	239	277	293	282	280	279	276	266	271	267	247	236	
	Edh at Noon	20	25	28	36	39	37	36	37	29	21	17	16	
All-Sky Solar Radiation	RadAvg	363	687	1126	1521	1866	1906	2049	1651	1203	719	366	262	
	RadStd	39	52	89	136	136	145	104	124	110	97	41	28	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (20 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page